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Final Project Formal Report
Optical Heart Rate Sensor

Abstract

The objective of this project is to successfully implement a working optical heart rate sensor using optical sensors, active filters and comparators to read and manipulate the signal read between a finger into a usable output. Throughout the project certain criteria like a range between 40 - 200 BPM, power limits, proper gain, cutoff frequency and others were monitored and adjusted to eventually output a clean pulse wave that can be used as an input to a LED.

The optical stage made use of a phototransistor and receiver, which was sent through filters which utilized LM324 and its output was then passed to a comparator to convert it from an analog to a digital output using a LM393. This output was then sent to an LED which was connected to an NMOS to ensure proper transmission and led to an accurate output.

Materials Used

IR204A, PT204-6B, LM324, LM339, RFD3055, LED, AD2, CAPACITORS, RESISTORS

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Introduction

In this project we will aim to implement knowledge and designs inherited from previous labs to create an optical heart rate sensor. An optical heart rate sensor is a commonly found application in smart watches, oximeters and various other medical fields. It works through a method known as photoplethysmography or PPG in short which finds change in blood volume within a body area by measuring the transmitted and reflected light. In this project we will use our finger to detect heartbeats similar to an oximeter. The **Optical Sensor** input starts from a phototransmitter which is read by a receiver with a finger in between. Since this input value read would be quite small in voltage with pulse rates between 40 and 200 beats per minute giving very small readings. These readings would have to be amplified through a large gain if we aim to see an usable voltage to indicate an LED in the end. This is where our **Active Filters** would help. By passing this input through a low pass filter and a high pass filter we can get a frequency range that is specific to the human body while providing it with a massive gain at the same time. The output from the filters would be an analog signal which can't really be used to give an accurate reading for our final stage. It needs to be converted to a stable digital pulse output. This is where a **Comparator** can be implemented to convert this output from analog to digital. The output from this comparator would be final and at this stage we would only need a LED to act as an **Indicator** which would complete this circuit. The report would be centered around these 4 stages separately and how they come together to work as an **Optical Heart Rate Sensor**.

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Theory

This circuit involves intricate designing of a lot of components. While the design of each component would be finalized at its respective stages there are some basic equations which can be derived which would help with these designs. This section will also cover an overview of how each stage works.

1- Optical Sensor

The sensor is the input detected through emitting an IR204 LED and read through a PT204-6B phototransistor with the object in between. If blood volume increases the amount of light passed through changes, changing the input. To find the LED current to ensure these components don't exceed operating limit we can use KVL:

$V_{cc} - I_D R_D - V_D = 0$

$I_D = (V_{cc} - V_D) / R_D$ (1)

$P = (V_{cc} - V_D)^2 / R_D$ (2)

To find make sure the output from the PT204-6B passes current as expected we can use the equation:

$V_{PT} = i_{PT} R_E$

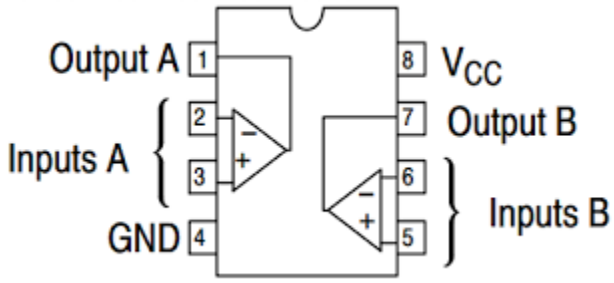
2- Active Filters

The filters contribute to a large part of the circuit. The incoming signal from the sensor is quite small and won't power any components. To find what frequency range is required we can use f = BPM/60. This would give our cutoffs for a human heartbeat range filtering out the other frequency through a high pass and a low pass filter creating a range. The cutoff frequency for each stage can be found through $f_c = 1/2\pi RC$ (3). The gain can be found V_{out} / V_{in} and using that we can derive $1 + R_f / R_1$ (4) which will be a large value needed to amplify the power

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3- Comparator and output

The comparator is based around the LM393 with its pinout:



The signal from the filters would still have a certain amount of noise and to counter this hysteresis is used to ensure only the correct signals are being read. The hysteresis voltage can be derived using:

$V_{Hyst} = (V_{DD} - V_{SS}) R_1 / R_1 + R_2$ (5)

This value would have fit in the range smaller than the peak 2 peak of the heartbeat amplitude but larger than the signal noise

For the LED that reads the output from this comparator we have to make sure it stays within a safety limit. This can be designed through

$I = (V_{DD} - V_{LED}) / R_L$

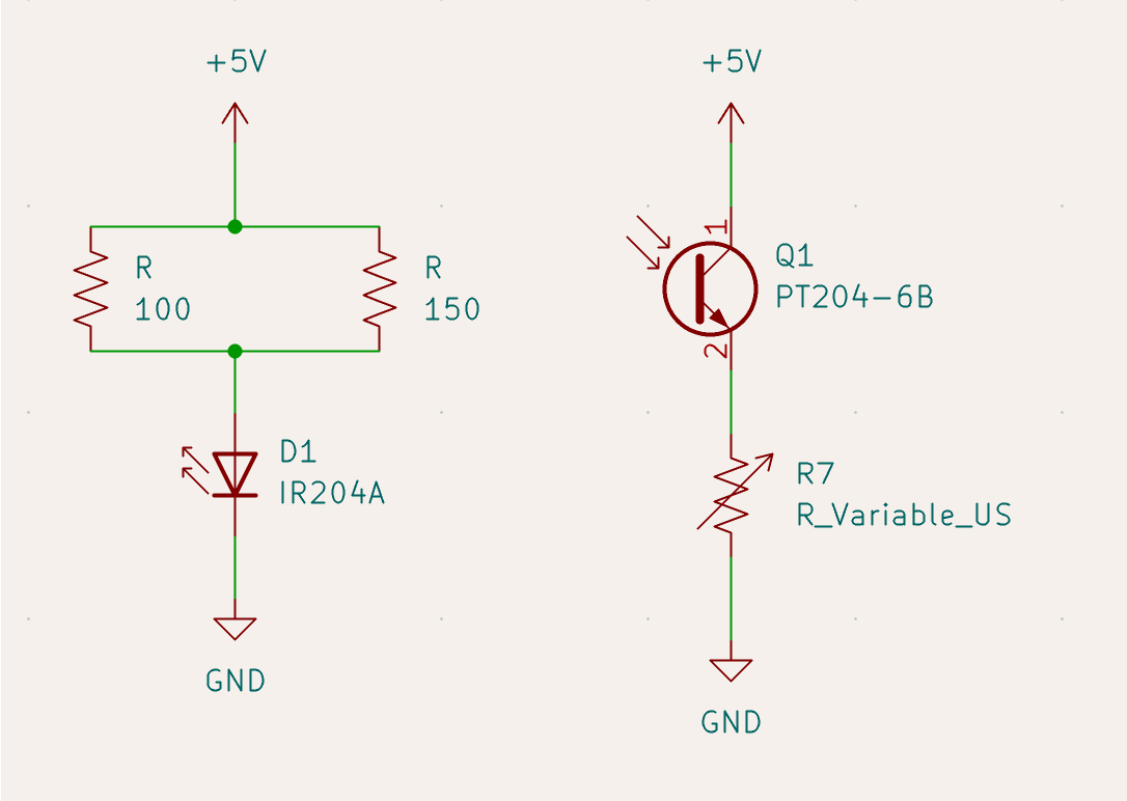
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Task 1 - Optical Sensor

Objective

This task will aim to design an optical sensor using a receiver and a transmitter to detect signals. It will later be used to detect change in blood volume between fingers

Procedure



1. Calculate the required resistor values
2. Replicate the circuit above
3. Capture screenshot without any interference between
4. Verify function of circuit by placing a finger in between and comparing outputs

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Results

Using equation (1) with Vcc as 5V and Max Value of IR204 as 1.5V (V_D used as 1.2V) at 20mA, Max I_D at 100mA, 0.25W for max power rating.

We used two resistors in parallel 100 ohms and 150 ohms to ensure each resistor didn't go over the power limit which gave an equivalent resistance of 60 which was used in calculations below

I_D for LED current = 63.3 mA (1)

P_{Total} = 0.24 W (2)

Our I_D is within the safe limit by being less than than the max 100 mA

For our R_E value we found it best to use a potentiometer with an upper limit of 10k ohms



Readings without a finger in between (Peak 2 peak - 14.4 mV, Frequency - 9.6 Mhz)

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Readings with a finger in between (Peak 2 Peak - 40.157 mV)

Conclusion

The optical sensor was a success as we could see visible change when a finger was placed in between. We can see a baseline voltage well above the $V_{CE,\ sat}$ of 0.4V which indicates the transistor operates linearly as expected. It is also visible to see small waves when the finger is placed showing that it is reading through it

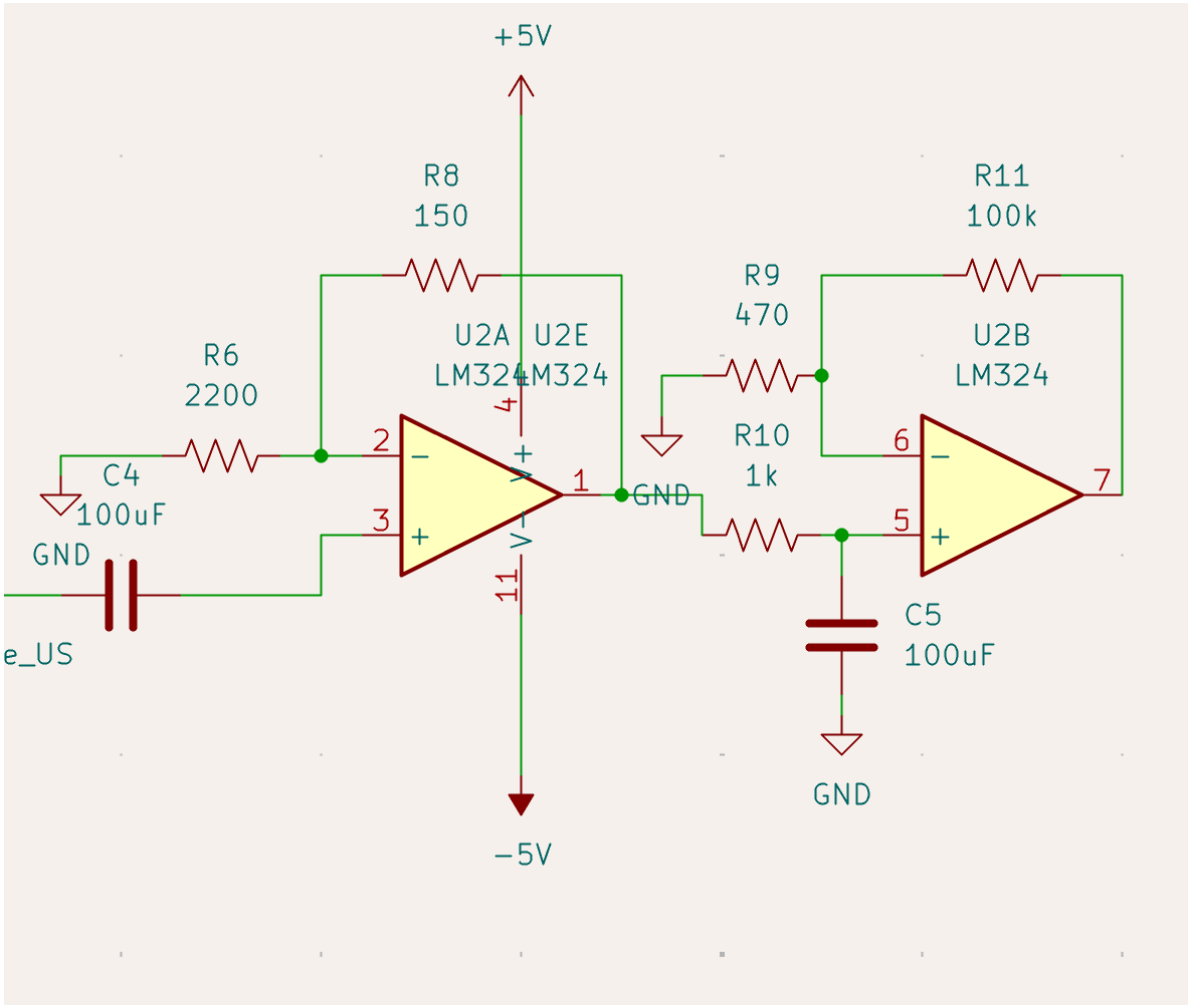
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Task 2 - Active Filters

Objective

This task aims to tune the heartbeat signal by giving it a range through cutoff frequencies and remove the DC offset all while providing a significant gain using a high pass and a low pass filter

Procedure



1. Replicate the circuit above

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2. Select circuit components that would ensure proper gain and cutoff frequency
3. Capture a FRA plot and other outputs

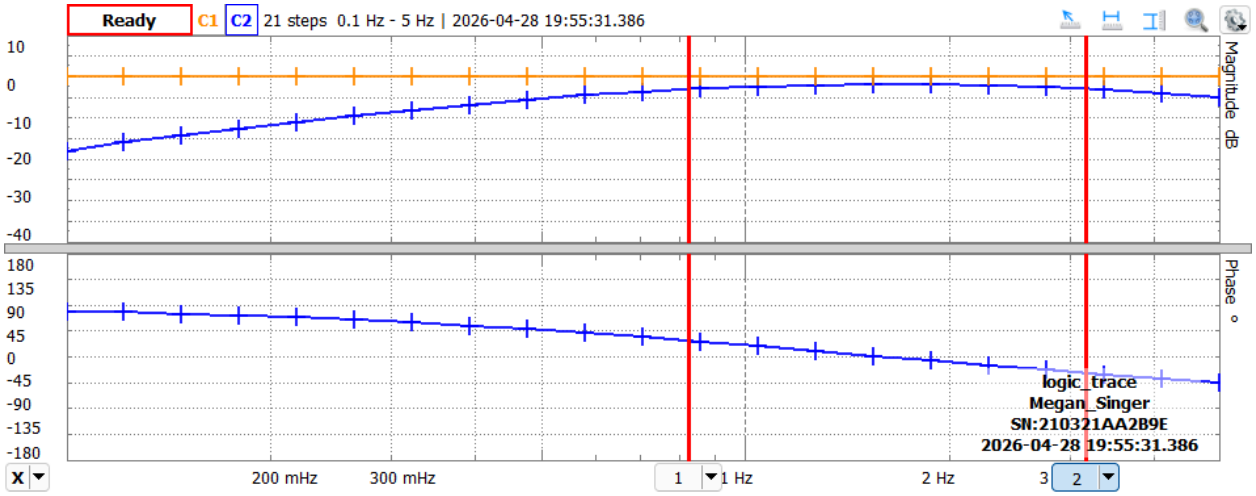
Results and Calculations

To get our design targets we find the lower range to by converting the BPM to frequency by dividing by 60 which gives us a range of 40/60 - 200/60 = 0.667 Hz - 3.33 Hz

For our high pass filters we picked the resistors 2.2k and 150 ohms in series and a capacitor of 100uF which using equation (3) our cutoff was found to be 0.677 Hz

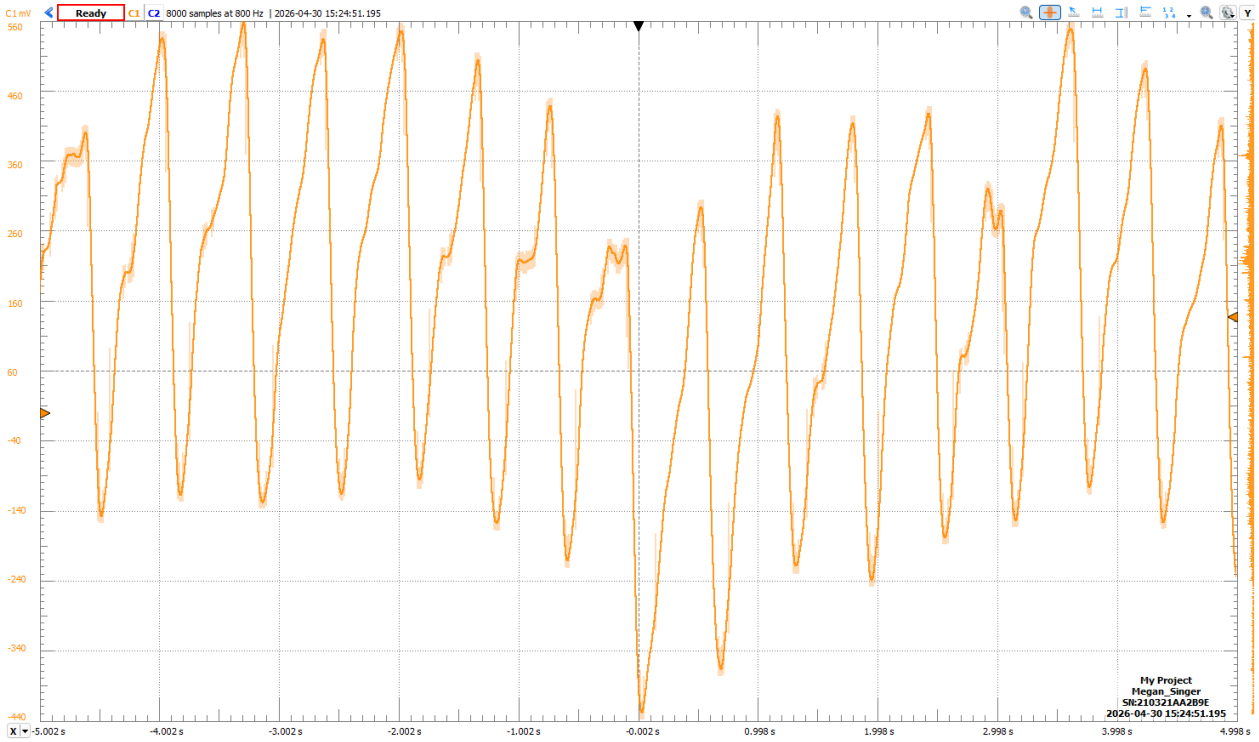
For our high pass filters we picked the resistor 470 ohms and a capacitor of 100uF which using equation (3) our cutoff was found to be 3.386 Hz

These values fit our range. Now for gain we used equation (4) with R_f = 100k ohms and R_1 = 1k ohms which gave a gain of 101 V/V which would translate our input to an usable output



FRA output, circuit follows cutoff as expected

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Oscilloscope output with finger in between sensors, peak 2 peak nearly 3V

Conclusion

This task was a success in implementing filters to achieve gain and create a range for cutoff frequency. We successfully outputted an analog signal with our sensor input. Our FRA further reinforced the accuracy by showing that the passband lies within our range. Our values can now be used to convert into a digital signal and finish the circuit

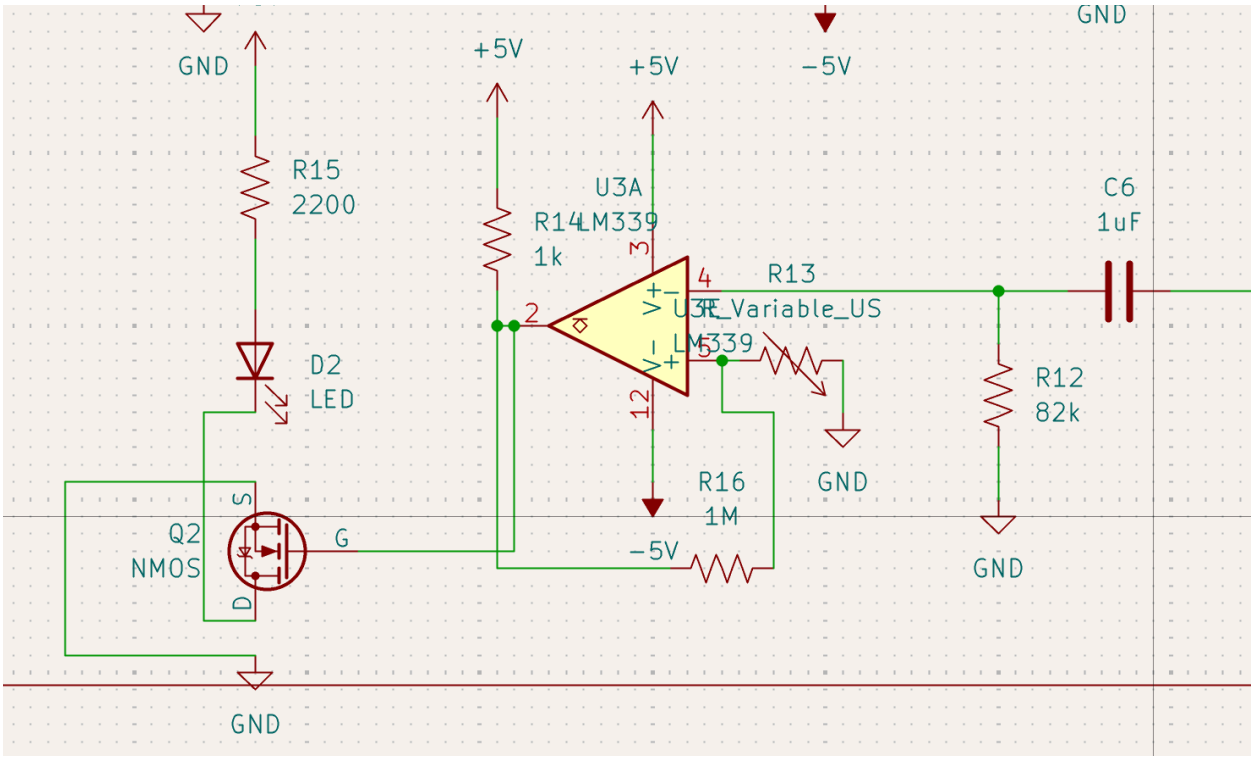
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Task 3 - Optical Heart Rate Sensor

Objective

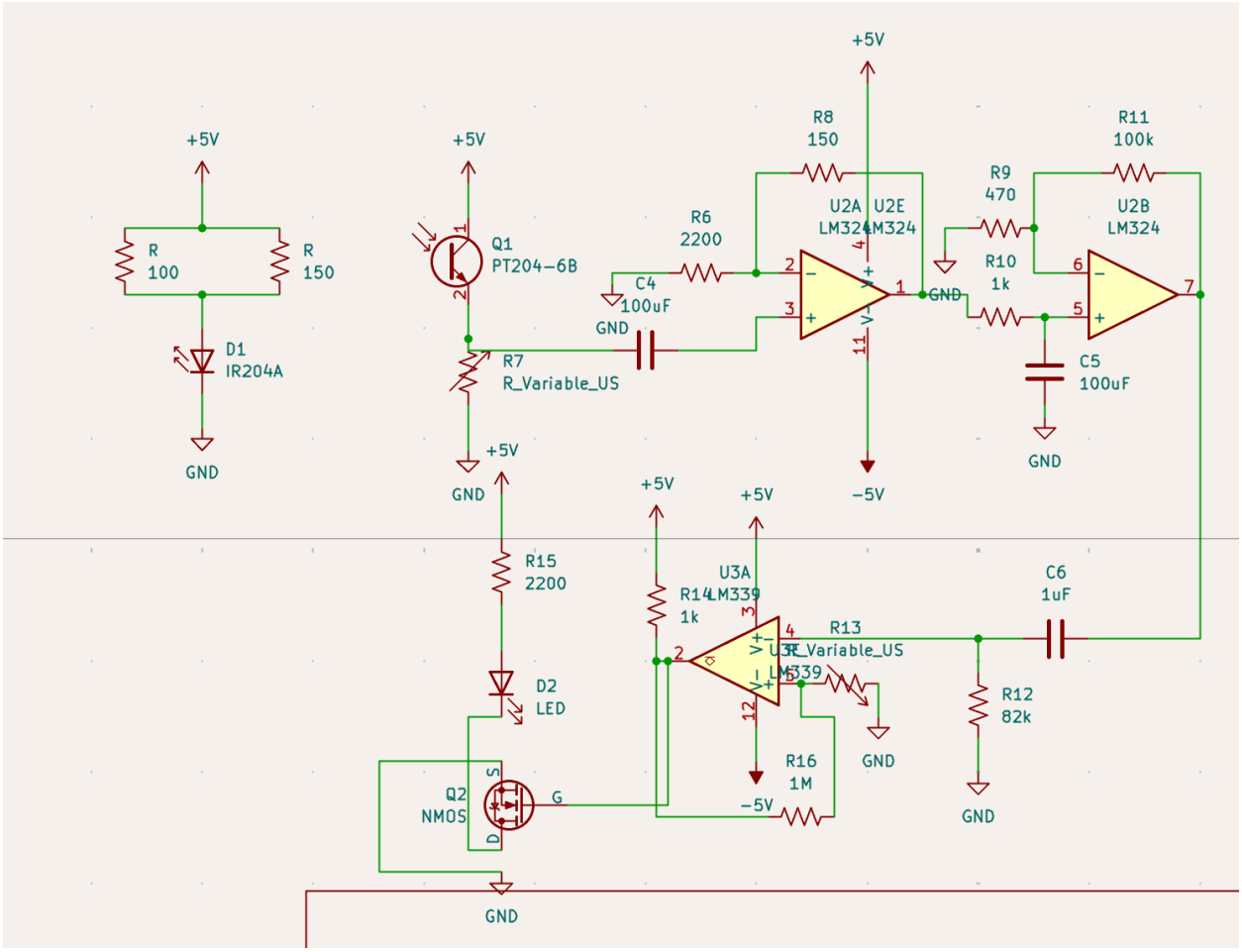
This task aims to build the comparator upon which its output should be the final output of the optical heart rate sensor. The comparator would change the signal to a digital signal which will be fed into our LED to show as an indicator which will blink in sync with our heartbeat

Procedure



1. Build the circuit above to implement a comparator and an indicator
2. Find resistor values required
3. Verify LED turns on

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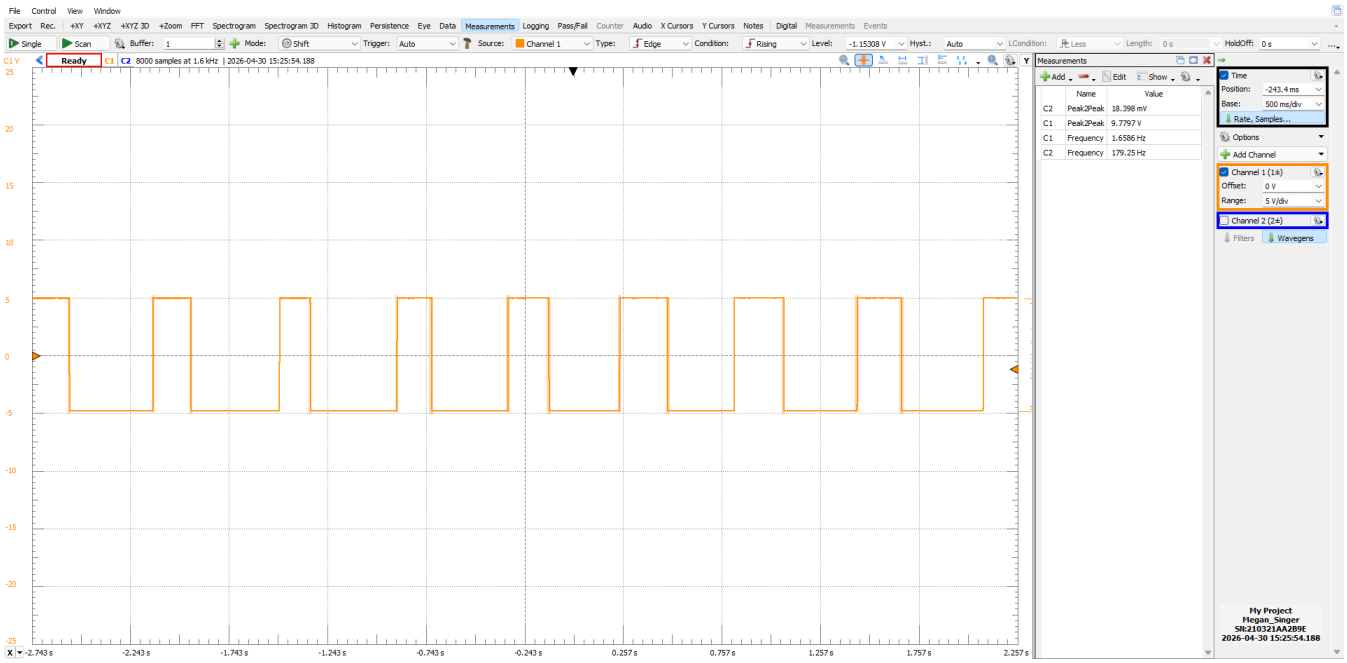
4. Make sure final circuit matches circuit above
5. Get an oscilloscope output to verify circuit function

Results and Calculations

To make sure only the amplified heartbeat is, we implemented hysteresis. We targeted a V_{hyst} value of 100mV. Using equation (5) and R1 = 10k, R2 = 1M, Vdd = 5, Vss = -5 we get V_{hyst} = 99mV

To also ensure our input frequency stays within range that would effectively capture pulse we use equation (3) with R = 82k and C - 1uF which gives us a cutoff frequency of 1.94 Hz which sits right between our range

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Final output - Frequency - 1.6586 Hz, Peak 2 Peak - 9.7797V

Conclusion

Overall this experiment was a success. From the optical sensor input we successfully converted it twice, once through the filters and that went through the comparator which gave us a stable square pulse wave very similar to that of a human pulse. Throughout the project there was the source of error of no way to block external noise like light with input which meant the final output isn't 1:1 to human heartbeat. However in the end our output was exactly as anticipated with a clean pulse showing that every subsystem was working as it was meant to leading to a proper output.